

**GEETHANJALI INSTITUTE OF SCIENCE & TECHNOLOGY****(AN AUTONOMOUS INSTITUTION)****(Approved by AICTE, New Delhi & Affiliated to JNTUA, Ananthapuramu)****(Accredited by NAAC with "A" Grade, NBA (EEE, ECE & ME) & ISO 9001:2008 Certified Institution)****QUESTION BANK (DESCRIPTIVE)****Subject Name: SOFTWARE ARCHITECTURE AND DESIGN PATTERN****Subject Code: 23A0538Ta****Course & Branch: B.Tech C S E****Year & Semester: IV B.Tech I Semester****Regulation: RG23****Unit 1**

S. No	2 Marks Questions (Short)	[BT Level] [CO] [Marks]
1.	Define Software Architecture.	L1, CO1, 2M
2.	What is Architecture Business Cycle?	L1, CO1, 2M
3.	What are Architectural Patterns?	L1, CO1, 2M
4.	Define Reference Model.	L1, CO1, 2M
5.	What is a Reference Architecture?	L1, CO1, 2M
6.	What are Architectural Views?	L1, CO1, 2M
7	Define Quality Attributes.	L1, CO2, 2M
8.	What is Achieving Qualities?	L1, CO2, 2M
9	What is Software Architecture Documentation?	L1, CO2, 2M
10	What is Reconstruction of Architecture?	L1, CO2, 2M

S. No	Descriptive Questions (Long)	[BT Level] [CO] [Marks]
1	Explain Software Architecture and its importance with example.	L2, CO1, 10M
2	Describe the Architecture Business Cycle in detail.	L2, CO1, 10M
3	Explain different architectural structures and views.	L2, CO1, 10M
4	Discuss architectural patterns with examples.	L2, CO1, 10M
5	Explain the concept of reconstructing software architecture	L2, CO1, 10M
6	Explain Quality Attributes with example	L2, CO2, 10M
7	Explain achieving qualities with following Tactics a) Availability Tactics b) Security Tactics c) Testing Tactics	L2, CO2, 10M
8	Describe the process of Designing Software Architecture.	L2, CO2, 10M
9	Explain various Architectural Styles with examples.	L2, CO2, 10M
10	Explain Documentation of software architecture.	L2, CO2, 10M

Unit-2

S. No	2MarksQuestions(Short)	[BT Level][CO][Marks]
1.	What is Architecture Evaluation?	L1,CO3,2M
2.	Define ATAM.	L1,CO3,2M
3.	Define CBAM.	L1,CO3,2M
4.	What is design decision making?	L1,CO3,2M
5.	What are Software Product Lines?	L1,CO3,2M
6.	Define Component-Based Systems.	L1,CO3,2M
7.	What is off-the-shelf component?	L1,CO3,2M
8.	What is system evolution?	L1,CO3,2M
9	What is architecture analysis?	L1,CO3,2M
10	Define interoperability.	L1,CO3,2M

S.No	Descriptive Questions(Long)	[BT Level] [CO][Marks]
1	Explain Architecture Evaluation techniques.	L2,CO3,10M
2	Describe ATAM method in detail.	L2,CO3,10M
3	Explain CBAM with example.	L2,CO3,10M
4	Discuss architecture design decision-making process.	L3,CO3,10M
5	Explain Software Product Lines with advantages.	L2,CO3,10M
6	Describe building systems from off-the-shelf components.	L2,CO3,10M
7	Discuss future trends in software architecture.	L2,CO3,10M
8	Compare and contrast ATAM and CBAM.	L2,CO3,10M
9	Explain any 2 Architecture Evaluation techniques with example	L2,CO3,10M
10	Write short notes on system scalability and evolution.	L2,CO3,10M

Unit-3

S. No	2MarksQuestions(Short)	[BT Level][CO][Marks]
1.	What is a Design Pattern?	L1,CO4,2M
2.	What is Pattern Catalog?	L1,CO4,2M
3.	Define Creational Pattern.	L1,CO4,2M
4.	Define Structural Pattern.	L1,CO4,2M
5.	What is Singleton Pattern?	L1,CO4,2M
6.	What is Factory Method?	L1,CO4,2M
7.	What is Adapter Pattern?	L1,CO4,2M
8.	What is Facade Pattern?	L1,CO4,2M
9	What is Composite Pattern?	L1,CO4,2M
10	What is Bridge Pattern?	L1,CO4,2M

S. No	Descriptive Questions(Long)	[BT Level][CO][Marks]
1	Explain Design Patterns and their role in software design.	L2,CO4,10M
2	Describe pattern catalog and its organization.	L2,CO4,10M
3	Explain Creational Patterns with examples.	L2,CO4,10M
4	Describe Factory patterns in detail.	L2,CO4,10M
5	Explain the intent , Collaboration, Motivation of Builder Pattern	L2,CO4,10M
6	Explain the intent , Collaboration, Motivation of Bridge Pattern	L2,CO4,10M
7	Explain the Abstract Factory with Example	L2,CO4,10M
8	Explain the Collaboration, Consequences and Sample code of Façade Pattern	L2,CO4,10M
9	Illustrate the intent , motivation, Applicability of Prototype Pattern	L4,CO4,10M
10	Explain the Collaboration, Consequences and Sample code of Composite Pattern	L2,CO4,10M

Unit-4

S. No	2MarksQuestions(Short)	[BT Level][CO][Marks]
1.	What is Behavioral Pattern?	L1,CO5,,2M
2.	Define Observer Pattern.	L1,CO5,2M
3.	What is Strategy Pattern?	L1,CO5,2M
4.	What is Command Pattern?	L1,CO5,2M
5.	Define Iterator Pattern.	L1,CO5,2M
6.	What is State Pattern?	L1,CO5,2M
7.	What is Mediator Pattern?	L1,CO5,2M
8.	What is Memento Pattern?	L1,CO5,2M
9	What is Template Method Pattern?	L1,CO5,2M
10	What is Visitor Pattern?	L1,CO5,2M

S.No	Descriptive Questions(Long)	[BT Level][CO][Marks]
1	Explain Behavioral Patterns with examples.	L2,CO5,10M
2	Explain structural Patterns with examples.	L2,CO5,10M
3	Explain the intent , Collaboration, Motivation of Strategy Pattern	L2,CO5,10M
4	Explain the intent , Collaboration, Motivation of State Pattern	L2,CO5,10M
5	Explain the motivation, structure, participants of Mediator Pattern	L2,CO5,10M
6	Describe Template Pattern with Example	L2,CO5,10M
7	Illustrate the Motivation, Structure and participants consequences of Memento Pattern	L4,CO5,10M
8	Explain the Motivation , Consequences and Sample code of Template Pattern	L2,CO5,10M
9	Explain the Collaboration structure and Participants of State Pattern	L2,CO5,10M
10	Explain the visitor Pattern with example	L2,CO5,10M

Unit-5

S.No	2MarksQuestions(Short)	[BTLevel][CO][Marks]
1	What is a Case Study in software architecture?	L1,CO6,2M
2	What is A-7E system?	L1,CO6,2M
3	What is World Wide Web architecture?	L1,CO6,2M
4	Define interoperability.	L1,CO6,2M
5	What is high availability?	L1,CO6,2M
6	What is Air Traffic Control system?	L1,CO6,2M
7	What is Celsius Tech system?	L1,CO6,2M
8	What is product line development?	L1,CO6,2M
9	What is distributed system?	L1,CO6,2M
10	Define system reliability.	L1,CO6,2M

S. No	Descriptive Questions(Long)	[BT Level][CO][Marks]
1	Compare and contrast the creational and structural patterns	L4,C06,10M
2	Compare Structural and Behavioral Pattern	L4,C06,10M
3	Explain A-7E case study in detail.	L2,C06,10M
4	Describe World Wide Web architecture.	L2,C06,10M
5	Discuss Air Traffic Control system architecture.	L2,C06,10M
6	Describe Celsius Tech product line development.	L2,C06,10M
7	Compare different case studies.	L4,C06,10M
8	Explain lessons learned from case studies.	L2,C06,10M

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